Notebook Companion

A study notebook you can just talk to, built on Gnani AI

Live notebook-companion.onrender.com
Code github.com/nagasaiparaksh/notebook-companion-v5
Demo demo-video/notebook-companion-voice.mp4 (3:35)

1 The problem

- Students write while the teacher talks, so they **listen less** and copy more.
- Indian classrooms are **spoken and code-mixed** (Hinglish and more), but note apps are English-first and typing-first.
- Generic speech tools **mishear course terms** ("are and ends" for RNNs).
- Notes stay **passive**: nothing quizzes you or shows what you missed.

2 The solution

A voice-first notebook: **tap the mic once and talk**. For every sentence the AI decides whether it is dictation, a command or a question, then writes, acts or answers.

Students upload their **syllabus or slides**; from then on recognition, answers, quizzes and "what did I miss?" all follow *their* course.

3 Everything by voice

"Plants make glucose from sunlight…" dictation · "Write a summary with a heading and a diagram" · "Change the heading to…" "Undo that" · "Explain my notes simply" spoken answer from your notes · "Add that to my notes" · "Quiz me" "Make flashcards" "What did I miss?" · "Record the lecture" live captions → organised notes · "Photosynthesis में plants…" Hindi + English in one sentence

4 Gnani APIs used

Prisma v3.0 STT (REST)	Dictation and commands; **keyterms, topic, previous_text** from the student's course material; code-mixed output
Prisma v3.0 Streaming STT	Live lecture captions over WebSocket, relayed by the server (key never in the browser), with 15-min session rollover
Prisma v3.0 Batch STT	Uploaded lecture recordings with **speaker diarization** (teacher vs students)
Timbre v3.0 TTS (REST + SSE)	Answers and read-aloud, streaming so speech starts in ~1 s; 13 voices, all supported languages

5 What makes it different

- **Syllabus-aware recognition:** course terms boost Gnani Prisma; a built-in A/B test shows the difference.
- **Code-mixed first:** English + 9 Indian languages, mixes of them, or auto-detect.
- **Acts, not just transcribes:** writes, edits, draws, quizzes, all undoable.
- **From passive to active recall:** every notebook becomes a quiz, flashcards and a gap check.
- **Private:** per-user accounts; export to PDF, Word or Markdown.

6 Market & impact (assumptions)

4.5 cr	24.7 cr	$29B	₹53 cr
higher-ed students	school students	India edtech by 2030	/yr at 1% × ₹99/mo

If 1% of India's college students paid ₹99/month, that is 4.5 lakh users ≈ ₹53 crore a year, before school and coaching students or college-wide licences. Voice-first also helps learners who write slowly or have motor or visual difficulties.

Sources: AISHE 2023-24; UDISE+ 2024-25; IAMAI–Grant Thornton ($7.5B in 2024 → ~$29B by 2030).

7 Built with

React + TypeScript + Vite · Python Flask · SQLite · WebSockets · Gemini LLM + automatic fallback · RAG over notes & syllabus · Docker on Render · 32 Playwright end-to-end tests

Next: Android app with offline drafts, a teacher view that shares a lecture's notes with the whole class, and syllabus packs for common university courses.

Built by Y Naga Sai Paraksh for the Gnani AI competition

Try it: sign up in 30 seconds at notebook-companion.onrender.com